

Market Research: Smart Electricity Bill Analysis Tool for Indian Households

Electricity Usage Patterns & Pain Points (Urban vs Rural)

- **Rapidly rising residential load:** Indian household electricity demand has surged in recent years (now $\approx 24\%$ of national use, up from 4% in 1971 ¹). TV ownership is near-universal ($\approx 95\%$), $\sim 60\%$ have refrigerators, while air conditioners remain concentrated in high-income urban homes (42% of top urban quintile) ². Even low-income rural homes often own TVs ($\approx 86\%$) and fans, but far fewer appliances (fridges, coolers) ³ ⁴. Grid-connected rural households consume significantly more (lighting, cooling, entertainment, appliances) than off-grid ones ⁴.
- **Reliability & outages:** Many consumers still face irregular supply. Urban areas get ≈ 22 hours/day on average vs ≈ 20 h in rural areas ⁵. About 76% of households report unanticipated outages (higher in rural than urban) ⁶, and half face supply-quality issues monthly (e.g. blackouts, low voltage) ⁶. Chronic outages are still common in some states (e.g. Bihar, Odisha). This unreliability is a major pain point, especially for rural users relying on agriculture or small enterprises.
- **Billing issues & costs:** Traditional fixed-charge tariffs have caused frustrations: discoms often cite non-payment and theft, but many consumers dislike high base fees. A recent survey found **59.5% of respondents prefer consumption-based billing with lower base charges** over flat fees ⁷. Users with many appliances (more typical in rural grid homes) strongly favor paying for what they use ⁷. Erroneous bills also plague consumers: for example, faulty meter readings have led to shock bills (news reports of exorbitant bills) and undermine trust. Cash payment is still the norm (only $\sim 17\%$ pay digitally ⁸), and many rural consumers get irregular or no bills ($\approx 6\text{--}10\%$ of grid households receive bills only sporadically or not at all ⁹), leading to confusion about consumption and charges.

Smart/Prepaid Meter Adoption & Awareness

- **Government rollout:** India's Smart Meter National Programme (part of the RDSS scheme launched 2017) aims to install **250 million smart/prepaid meters by 2025-27** ¹⁰. This is a massive effort ($\text{₹}97,631$ crore budget ¹¹), led by EESL under the Power Ministry. By October 2024, ~ 117.7 million smart meters had been **awarded (contracts let)** nationwide, of which only ~ 14.5 million were actually installed ¹⁰. Prepaid meters are a core focus: as of early 2025, ~ 19.79 crore (198 million) prepaid smart meters were sanctioned under RDSS ¹². States like Uttar Pradesh, Maharashtra, Tamil Nadu are leading in roll-out; some, like Kerala, initially joined the Central program but later sought their own model ¹³ ¹⁴. Despite the ambitious targets, actual deployment lags (e.g. only ~ 5.5 M installed by early 2023 ¹⁵).
- **Consumer response:** Among those with smart/prepaid meters, feedback is mixed but generally positive. A 2023 CEEW survey (2,700 urban users) found **$\approx 60\%$ of smart meter consumers were satisfied** with the technology and billing, 92% had smooth installations, and $\sim 50\%$ saw more regular billing after smart meters ¹⁶. Over 60% of respondents would recommend prepaid smart meters to friends/relatives ¹⁶, indicating willingness to endorse the new system. Prepaid users report benefits like a sense of control over expenses, reduced theft and improved local supply ¹⁷. Uptake of

supporting mobile apps is however low: only ~50% of users are even aware of the smart meter app, and just 45% use it ¹⁸. Most consumers (~70%) still prefer receiving paper bills or SMS summaries (SMS often lacks breakdown) rather than relying solely on digital apps ¹⁸. Regional differences are stark – for example, in Bihar 80% of users used the smart-meter app (thanks to strong outreach), whereas in Haryana/MP it was ~14% ¹⁸.

- **Digital literacy gap:** High-end smartphone penetration (~70% of households ¹⁹ and ~425M rural users ²⁰) means the potential audience is large. But digital adoption in utilities remains low: only ~17% of households pay electricity bills online (27% urban, 12% rural) ⁸, reflecting trust or convenience issues. Thus, while smart meters empower consumers with data, actual awareness of their features (especially apps) is limited ¹⁸. Consumers need education on the benefits (as ministers noted, prepaid meters should reduce costs by ~2–2.5% ²¹) to drive adoption.

Existing Software-Only Tools & Apps

- **State utility apps:** Some distribution companies offer apps for prepaid customers. For instance, Bihar's "Sugam Smart Meter" app lets prepaid users track *real-time usage*, *remaining balance*, and even **predict how many days of supply** their credit will last ²². It also provides usage breakdown (daily/weekly/monthly) and alerts for low balance ²². Such apps integrate payment (recharge) and complaint features. However, consumer reviews highlight functional issues (validation failures, customer service delays) ²³, indicating user experience is often poor.
- **Third-party/AI apps:** A few tech startups offer software solutions that work via smartphone without extra hardware. For example, **Bharat Smart Services (BSS)** provides a free "Bijli Auditor" app where consumers photograph their (analog) meter; AI then extracts the reading and tracks usage. BSS reports that "*lakhs of consumers track their daily usage, reduce wasteful consumption, and lower their bills*" using its app ²⁴. (In practice, the app combines AI meter-reading with analytics/tips.) This kind of app is fully software-based – the only input needed is the meter image or data.
- **Global analogs:** Internationally, numerous "green energy" apps exist. For instance, the US "Meter Readings" app lets users **manually input multiple meter readings** (electricity, gas, etc.) and then displays usage, cost and trends ²⁵. Similarly, "Energy Consumption Analyzer" (Android) lets users log various meter readings and view usage graphs ²⁶. These apps can even forecast a bill based on past rates. Others like "Green Outlet" or "Energy Cost Calculator" help users estimate appliance costs by entering usage patterns ²⁷ ²⁸. While most of these assume foreign tariffs, they show that pure-software tools (often free) can engage users by providing consumption insights without any new hardware.

Gaps & Unmet Needs in Energy Monitoring Solutions

- **Usability & literacy:** Current apps often assume literate users. Many electricity consumers (especially the elderly or semi-literate) struggle with text-heavy interfaces. Indeed, half of smart-meter users were unaware of apps ¹⁸, and 70% still want paper bills. There is a lack of localized, intuitive interfaces: few offer regional languages or voice guidance. Semi-literate users need icon-driven, visual cues and very simple navigation (as design studies for seniors recommend simplifying layouts and using large buttons ²⁹ ³⁰). Without such adaptations, a large segment cannot effectively use digital tools.
- **Accessibility & inclusion:** Beyond literacy, the special needs of elderly or disabled users are often ignored. For example, design guidelines stress the importance of *enlarged text/buttons and high contrast* for older eyes ²⁹ ³⁰. But most energy apps lack such features. There is also virtually no

support for sight-impaired (e.g. voice-over compatibility) or motor-impaired (e.g. one-click actions) users. The lack of audio prompts and difficulty with mini-keyboards mean many users simply cannot interact with these apps comfortably.

- **Data access & trust:** Consumers typically do not have easy access to detailed usage data. Smart meters generate granular data, but consumers can't readily retrieve it. Regulatory work (Prayas) recommends giving consumers right to their own data and accessible portals ³¹, but this is not yet implemented widely. Without transparent data, consumers cannot analyze their patterns or trust a third-party tool. Privacy is another concern: high-resolution data can reveal household habits, so robust data protection must be ensured.
- **Offline needs & connectivity:** Many rural areas lack reliable internet. Current solutions rarely work offline; and even recharging prepaid meters may fail if networks are down ³². For instance, remote disconnections (when credit runs out) can be catastrophic at night or in remote villages if people can't top up in time ³³ ³². Any tool must handle intermittent connectivity—possibly by working via SMS or by letting users enter meter readings offline for later sync.
- **Customer support & trust:** Trust is low when tools fail. Prepaid meters allow disconnection for non-payment, causing consumer anxiety ³³. There have been many complaints about app failures: e.g. Bihar users report bugs in the official prepaid app, payment errors, and unhelpful support ²³ ³⁴. Gaps like these (no guarantee of timely recharge or disconnection notice) create resistance. A successful solution must assure users of reliability, possibly with fallback options like SMS alerts or human assistance channels.

Willingness to Use Web/Mobile Tools

- **Large potential user base:** Smartphone ownership is now widespread in India (≈ 700 million users as of early 2023 ²⁰), including 425M in rural areas. Over half the population uses mobile internet ²⁰. This suggests a big potential audience for a digital tool. However, general adoption of energy apps has been slow: only $\sim 17\%$ of households use digital billing ⁸ (12% rural), meaning habit and trust barriers exist. Early indications are mixed: while smart-meter adopters express high satisfaction and willingness to recommend the technology ¹⁶, many still prefer familiar methods (cash payments, paper bills).
- **Cost-saving motivation:** With rising electricity tariffs and unreliable supply, consumers are highly motivated to save. Studies (e.g. Bain, 2024) show Indian consumers are increasingly conscious of sustainability and cost. If a tool clearly shows ways to cut bills or avoid surprises, users are likely to engage. CEEW found that prepaid-meter users experienced a **sense of control** and often **lower bills** after switching ¹⁷; messaging around these benefits could drive adoption. Early adopters (urban tech-savvy) already use “bill-saver” apps (as seen in college and tech communities). However, the elderly and rural users will need extra encouragement and possibly training or incentives.
- **Digital literacy campaigns:** Initiatives like rural smartphone training (for Anganwadi workers, etc.) show that even initially resistant users can learn digital skills ³⁵. Government pushes (e.g. linking smart meters to Aadhaar-style ID, mandating prepaid) indicate intent to normalize these tools. If rolled out with consumer education (bill clinics, customer care), and if the tool is extremely easy to use, willingness will grow. But the gap between smartphone ownership (70%) and digital bill payment (17%) ⁸ signals that any app must be **frictionless and trustworthy** to win users.

Accessibility for Elderly and Special-Needs Users

- **Simplified UI:** Design must follow senior-friendly guidelines. Experts stress *simplicity above all*: only essential functions, minimal steps, and a flat navigation. Key rules include “**simplify**” and “**increase size and spacing**” of buttons/controls ²⁹. Use large, high-contrast fonts (e.g. at least 20–30 pt for primary text ³⁶) and avoid small on-screen targets. Each function (e.g. “Log meter reading”, “View bill”) should have a big icon or button with a clear text label ³⁰. Instructions and labels should be kept extremely short ³⁷, using plain language.
- **Pictorial & audio cues:** For semi-literate users, include meaningful icons/graphics. Every icon/button should be accompanied by a simple text label or voice prompt ³⁰. For instance, pictures of a lightbulb for electricity, a wallet for costs. Consider voice-guided prompts or a read-aloud feature so users can hear meter readings, balances, or tips. Similarly, text alternatives (for visually impaired) and video/video-tutorial help (for those who struggle with text) can enhance inclusion. Mobile OS screen-readers (TalkBack, VoiceOver) should work with the app’s UI elements.
- **Regional languages:** The app must support Hindi and other local languages. Many rural elders do not read English; a Hindi (or vernacular) interface with voice prompts would make it accessible. Even support via WhatsApp or SMS in local language could be offered for key alerts (e.g. low balance warnings), since 70% of rural users have feature phones/smartphones ¹⁹.
- **Offline usability:** Given patchy internet in villages, basic functions (entering meter readings, viewing past bill details) should work offline. When online, the app can sync and fetch tariff rates or usage history. Offline modes and auto-fetch via SMS (where feasible) will reduce dependence on continuous connectivity.
- **Assistive features:** Include “accessibility settings”: adjustable text size, color themes (high contrast), and simple gestures (e.g. large swipe zones). Allow alternative inputs (e.g. numeric keypad for reading input, dictation). Provide a customer help/chatbot in local language. These accommodations help elderly or disabled users who may have motor, visual or cognitive limitations.

Sources: Authoritative surveys, government reports and recent studies were used. Key findings are cited from CEEW’s IRES 2020 study ⁵ ⁶, appliance ownership surveys ², smart meter policy analyses ¹⁰ ¹², and field research on smart meter users ¹⁶ ¹⁸. Global app examples are referenced via industry publications ²⁵ ²⁶. Design recommendations for elderly users come from usability research ²⁹ ³⁰. All sources are recent (2020–2024) to reflect the current Indian market.

¹ Microsoft Word - 1-0376_0549_001167-Tathagat.docx

https://www.clasp.ngo/wp-content/uploads/2021/04/Unraveling-urban-household-energy-use-patterns-in-India_ACEEE-2020.pdf

² ³ Appliance ownership trends in India: As per Consumer Pyramids Household Survey Data

<https://energy.prayasapune.org/power-perspectives/appliance-ownership-trends-in-india>

⁴ Drivers of Household Appliance Usage: Evidence From Rural India by Jennifer L Richmond, Shalu Agrawal, Johannes Urpelainen :: SSRN

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3605021

⁵ ⁶ ⁸ ⁹ ¹⁹ ceew.in

<https://www.ceew.in/sites/default/files/ceew-research-on-state-of-electricity-access-and-coverage-in-india.pdf>

- 7 Understanding electricity billing preferences in rural and urban India: Evidence from a conjoint experiment
<https://ideas.repec.org/a/eee/eneeco/v106y2022ics0140988321005831.html>
- 10 11 India targets 250 million smart meters by 2027, \$20 billion opportunity in energy management, ETEnergyworld
<https://energy.economictimes.indiatimes.com/news/power/india-targets-250-million-smart-meters-by-2027-20-billion-opportunity-in-energy-management/114162408>
- 12 sansad.in
https://sansad.in/getFile/loksabhaquestions/annex/184/AU481_n2Km9D.pdf?source=pqals
- 13 14 Power play: Kerala govt to scrap Centre's prepaid smart meter project | Thiruvananthapuram News - The Indian Express
<https://indianexpress.com/article/cities/thiruvananthapuram/power-play-kerala-govt-to-scrap-centres-prepaid-smart-meter-project-8909896/>
- 15 16 17 18 21 60% Smart Meter Consumers are Satisfied With the Technology, can pay Bills | CEEW
<https://www.ceew.in/press-releases/60-per-cent-smart-meter-consumers-satisfied-with-technology-can-pay-bills>
- 20 35 Smartphones boost digital literacy in India's rural daycares | World Economic Forum
<https://www.weforum.org/stories/2023/12/how-smartphones-can-boost-digital-literacy-among-indias-rural-communities/>
- 22 23 Bihar Sugam Smart Meter - Apps on Google Play
https://play.google.com/store/apps/details?id=com.mysugam.nbpdc&hl=en_US
- 24 Bharat Smart Services
<https://www.bharatsmartservices.com/>
- 25 26 27 28 15 "Green Apps" to Track Energy Usage — Electric Choice
<https://www.electricchoice.com/blog/green-apps-track-energy-usage/>
- 29 30 36 37 JMIR mHealth and uHealth - Design Guidelines of Mobile Apps for Older Adults: Systematic Review and Thematic Analysis
<https://mhealth.jmir.org/2023/1/e43186>
- 31 32 33 34 Smart metering of electricity consumers in India: getting it right
<https://energy.prayaspune.org/power-perspectives/smart-metering-of-electricity-consumers-in-india-getting-it-right>